Limits Graphically and Numerically

Section 1: Estimating Limits Numerically & Tables

1 A function g satisfies $g(1.9) = 3.81, g(1.99) = 3.98, g(2.01) = 4.02,$ and $g(2.1) = 4.19$. What is the estimated value of $\lim_{x \to 2} g(x)$?

a. 3.9

b. 4.0

c. 4.1

d. Does not exist

2 Based on table values of $f(x) = \frac{\sin(3x)}{x}$ as $x \to 0$ (x in radians), what is the estimated value of $\lim_{x \to 0} f(x)$?

a. 0

b. 1

c. 3

d. ∞

3 Values of a function $h(x)$ near $x = 3$ are $h(2.99) = 5.01$ and $h(3.01) = -4.99$. What does this suggest about $\lim_{x \to 3} h(x)$?

a. The limit is 0

b. The limit does not exist because left and right limits disagree

c. The limit is 5

d. The function is continuous at $x = 3$

Section 2: Graphical Analysis & Removable Discontinuities

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4 The graph of $f(x) = \frac{x^2-1}{x-1}$ has a hole at $(1, 2)$. What is the value of $\lim_{x\to 1} f(x)$?

a. 1

b. 2

c. 0

d. Undefined

5 A function f satisfies $\lim_{x\to 4^-} f(x) = 7$, $\lim_{x\to 4^+} f(x) = 7$, and $f(4) = 2$. What is $\lim_{x\to 4} f(x)$?

a. 2

b. 7

c. 4.5

d. Does not exist

6 Which of the following functions has a vertical asymptote at $x = 2$ and NOT a removable hole?

a. $f(x) = \frac{x-2}{x^2-4}$

b. $f(x) = \frac{x+1}{(x-2)^2}$

c. $f(x) = \frac{x^2-4}{x-2}$

d. $f(x) = \frac{2x-4}{x-2}$

Section 3: One-Sided Limits & Absolute Value Quotients

7 Evaluate the one-sided limit: $\lim_{x\to 0^+} \frac{x}{|x|}$.

a. 1

b. -1

c. 0

d. Does not exist

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8 Evaluate: $\lim_{x \to 5^-} \frac{x-5}{|x-5|}$.

a. 1

b. -1

c. 0

d. Does not exist

9 What is $\lim_{x \to 2} \frac{|x-2|}{x-2}$?

a. 1

b. -1

c. 0

d. Does not exist

Section 4: Infinite Limits & Horizontal Asymptotes

10 Evaluate: $\lim_{x \to 3^+} \frac{1}{x-3}$.

a. ∞

b. $-\infty$

c. 0

d. 1

11 Evaluate the limit at infinity: $\lim_{x \to \infty} \frac{7x^3 - 4x + 1}{2x^3 + 5x^2 - 3}$.

a. $\frac{7}{2}$

b. 0

c. ∞

d. $-\frac{1}{3}$

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12 What are all horizontal asymptotes of the function $f(x) = \frac{3x}{\sqrt{x^2 + 4}}$?

a. $y = 3$ and $y = -3$

b. $y = 3$ only

c. $y = 0$

d. No horizontal asymptotes

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